

DS Fact Sheet on Chemical Storage-Flammable and Corrosive Cabinets

National Institutes of Health (NIH) • Office of Research Services (ORS) • Division of Safety (DS)

Laboratory Safety and Chemical Storage

Limited laboratory space presents a challenge for the proper and safe storage of hazardous chemicals. Chemical segregation is critical for preventing unintentional chemical reactions between different hazard classes. Incompatible chemicals (i.e., those chemicals that can produce an undesired reaction if mixed) must be segregated and stored separately to reduce the risk of accidental mixing that can result in dangerous or violent reactions.

Chemical storage cabinets, such as corrosive and flammable cabinets, play critical roles in chemical segregation and lab safety. These cabinets are designed for the specific storage of different hazard classes and should not be used for multi-purpose or general chemical storage. Details of both cabinet types and their applicability are discussed below.

How to Use Corrosive and Flammable Cabinets?

Corrosive Chemical Cabinet	Flammable Cabinet
Purpose: • Provide designated storage for corrosive chemicals. • Vented exhaust, usually through a chemical fume hood to remove harmful vapors.	Purpose: • Flammable cabinets are designed and built to protect the contents from catching fire in case of a fire occurring outside and limiting any fire/explosion inside the cabinet from spreading to the lab. • Cabinets may come with vents; these vents should remain plugged to contain hazardous vapors within the cabinet.
What to store in a Corrosive Cabinet: • Corrosive materials - Strong acids and bases must be stored in separate cabinets, space permitting, when utilizing large quantities. - Small quantities of strong acids and bases can be stored inside the same cabinet, placed within secondary containment, and on separate shelves. - Within a designated storage area, separate organic acids from inorganic acids and organic bases from inorganic bases using appropriate secondary containment. - Cabinet doors must remain closed to keep vapors exhausting through the vent.	What to store in a Flammable Cabinet: • Flammable Chemicals - Any amount of class 1, 2 or 3 flammables over 500 ml/g and non-working stock must be secured in an approved flammable cabinet. - Do not exceed the listed storage capacity of the cabinet. - Keep cabinet doors always closed and secured to ensure proper functioning of the cabinet. - Flammable chemicals requiring refrigeration must be stored in refrigerators/freezers that are explosion proof and appropriately rated for the storage of flammable materials.
Find additional information on construction requirements, installation, and location at NIH DTR Corrosive Storage Cabinet	Find additional information on construction requirements, installation, and location at NIH DTR Technical Bulletin Flammable Storage Cabinets

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Additional Considerations for Chemical Segregation

- For chemicals with multiple hazards/multiple pictograms on the label (e.g., both flammable and corrosive), address the more severe hazard and store accordingly (flammable cabinet with separate secondary containment). OSHA details the GHS Pictograms and associated hazards on the [Hazard Communication Standard Pictogram Fact Card](#).
- Consider having an annual chemical clean-up or inventory assessment to check the accuracy of inventory, condition of containers, labels, expiration dates, and confirm that chemicals are properly segregated.
- Containers that are cloudy, have illegible labels, or if there is noticeable buildup around the cap should be reviewed for disposal. Outer containers that are rusted or otherwise show corrosion should be reviewed and if the chemical inside of the container is still usable, the outer container should be discarded. Dispose all older chemical that are no longer in use.
- Secondary containment can be used to separate hazard classes and to contain spills/drips to prevent accidental mixing of incompatible chemicals.



Above (left) – chemicals in secondary containment



Above (right) – spill trays to contain leaks or drips

- Secondary containment should be made of a non-reactive material, (e.g., Nalgene, polypropylene plastic, or an equivalent), and be clearly labeled with the hazard class of the contained chemicals.
- Hazardous chemicals must be kept at or below eye level and are not to be stored under the sink.

Additional information on working safely with chemicals can be found in the [NIH Chemical Safety Guide](#).